20-DAY LIVE CLASS AGENDA

Deep Learning • Attention • Transformers • LLM Fundamentals • RAG

Live Classes: 2 Hours Per Day

Recorded courses available in the portal: Python Programming, Advanced Python and Machine Learning.

Live class topics: Deep Learning, ANN, CNN, RNN, LSTM, GRU, NLP, Attention Mechanisms, Transformers, LLM Fundamentals and RAG.

Day 1 – Deep Learning

- Deep Learning overview
- Artificial Neural Network (ANN)
- Neurons, weights, biases and activation functions
- Forward propagation and loss functions

Day 2 – ANN Training

- Backpropagation
- Gradient descent and learning rate
- Optimizers: SGD and Adam
- Epochs, batches, overfitting and validation

Day 3 – CNN

- Image tensors and channels
- Convolution filters and feature maps
- Stride, padding and pooling
- Flattening and CNN classification

Day 4 – RNN, LSTM and GRU

- Recurrent Neural Networks (RNN)
- Hidden states and sequence processing
- Vanishing-gradient problem
- LSTM gates and cell state
- GRU architecture and comparison with LSTM

Day 5 – NLP and Sequence-to-Sequence

- NLP pipeline and text preprocessing
- Tokenization and vocabulary
- Sequence-to-Sequence problems
- Encoder–Decoder architecture
- Training and inference

Day 6 – Attention Basics

- Why attention is needed
- Attention intuition
- Query, Key and Value
- Attention scores
- Softmax weights and weighted-sum output

Day 7 – Scaled Dot-Product Attention

- Dot-product attention
- Scaling by the square root of key dimension
- Self-attention
- Attention masks and tensor dimensions

Day 8 – Multi-Head and Causal Attention

- Multi-head attention
- Parallel attention heads and concatenation
- Causal / masked attention
- Padding masks versus causal masks

Day 9 – Transformer Architecture

- Transformer architecture overview
- Encoder and decoder blocks
- Self-attention inside Transformers
- Cross-attention

Day 10 – Transformer Components

- Positional encoding
- Feed-forward network
- Layer normalization
- Residual connections
- Transformer training flow
- Encoder-only versus decoder-only models

Day 11 – BERT and Transformer Models

- BERT architecture and bidirectional context
- Masked language modelling
- Encoder-only model use cases
- Decoder-only model use cases
- BERT versus generative models

Day 12 – LLM Fundamentals

- What is a Large Language Model (LLM)?
- Tokenization
- Tokens versus words
- Embeddings
- Context window

Day 13 – LLM Training and Generation

- Pre-training
- Fine-tuning
- Instruction tuning
- RLHF / preference optimization
- Temperature, Top-k and Top-p

Day 14 – LLM APIs and Outputs

- API authentication and safe key handling
- System, User and Assistant messages
- Structured output
- Hallucination
- Prompt design and response validation

Day 15 – RAG Basics

- What is Retrieval-Augmented Generation (RAG)?
- RAG architecture
- Document loading
- Document parsing and text extraction
- Document cleanup

Day 16 – Chunking and Embeddings

- Chunking strategies
- Chunk overlap
- Embedding models
- Vector databases
- Metadata

Day 17 – Retrieval

- Similarity search
- Top-K retrieval
- Metadata filtering
- Retrieval failure modes

- Reranking

Day 18 – RAG Prompting and Citations

- Context construction
- RAG prompting
- Grounding answers in retrieved content
- Citations and source attribution
- Handling questions with insufficient evidence

Day 19 – RAG Evaluation

- Evaluation dataset and test questions
- Retrieval quality versus answer quality
- Faithfulness and relevance
- Citation checks
- Debugging and end-to-end integration

Day 20 – Project and Revision

- Build / demonstrate a document Q&A; RAG application
- Revision of Deep Learning, Transformers and LLMs
- Debugging common implementation issues
- Project presentation and feedback
- Introduction to deployment, monitoring and advanced RAG

PROJECTS

The following projects are included as practical project options across the course topics.

ANN Projects

- Stock Prediction System using ANN
- Indian Currency Note Classifier using ANN

CNN and Computer Vision Projects

- Handwriting Recognition Tool using CNN
- Virtual Mouse using OpenCV

RNN and NLP Projects

- Automatic Music Generation System using RNN
- Language Classifier

Generative AI Projects

- Text-to-SQL Analyst with Guardrails
- Hallucination Highlighter for RAG

- Code Agent